decisions, it will not live long and cannot pass on its genes. The overall balance between the evolutionarily stable strategies of all the animals in an ecosystem is called the evolutionarily stable state.

Canadian ecologist Crawford Stanley Holling introduced the idea of resilience—how ecosystems persist in the wake of disruptive changes such as fire, flood, or deforestation. A system's resilience can be seen in its capacity to absorb disturbance, or the time it takes to return to a state of equilibrium after a trauma. Ecologists now understand that ecosystems can have more than one stable state, and that resilient systems are not always good for biodiversity.

As the populations of many species are declining or becoming locally extinct, ecologists are once more focusing their attention on ecosystem resilience. Many, including French ecologist Michel Loreau, believe that if diversity in an ecosystem is reduced, the whole system will be less likely to resist major impacts such as the effects of climate change. Today, Loreau and others are working toward finding a new general theory that can explain the relationship between ecosystem biodiversity and resilience in order to understand and combat the effects of today's environmental challenges.